

CLASS XI · LECTURE 8

The Internet, Security and Viruses

One to two periods · Protocols, full forms, viruses and antivirus names

By the end of this lecture you can

- Define Internet, intranet, protocol, URL and browser.
- Name the protocol used for the web, for email and for file transfer.
- Define a virus, distinguish it from a worm and a Trojan, and name five antivirus products.
- Expand the short forms that appear in Section B.

The Internet is a network of networks

The **Internet** is a worldwide public network of networks that uses the TCP/IP family of rules. An **intranet** is a private network inside one organisation that uses the same kinds of tools (pages, browsers) but is not open to the public. A college notice site that only works on the campus LAN is an intranet. The public web is part of the Internet.

DEFINITION Protocol

A protocol is a set of rules that decides the format, timing and error handling of communication, so that two different computers can understand each other.

Name	Full form	The one fact to write
HTTP	Hypertext Transfer Protocol	Fetches web pages
HTTPS	HTTP Secure	The same, with encryption
HTML	Hypertext Markup Language	The markup language used to write web pages. It is not a protocol
SMTP	Simple Mail Transfer Protocol	The common protocol for sending email
FTP	File Transfer Protocol	Copies files between computers
TCP/IP	Transmission Control Protocol / Internet Protocol	The pair of rules the Internet runs on

The question “the most common protocol used for email” is **SMTP**. The question “markup language used to produce web pages” is **HTML**. HTML describes the page. HTTP carries the page. They are not substitutes.

A **browser** is application software that requests pages and displays them. A **URL** is the address of a resource, such as <https://www.biek.edu.pk>. An **ISP** is an Internet service provider, the organisation that connects you. You do not need the internal life of a search engine.

Viruses and the programs that stop them**DEFINITION Computer virus**

A computer virus is a program that attaches itself to another program or file, spreads when that host is copied or run, and may damage data, waste resources or open a door for misuse.

A virus is **software**. It is not a hardware fault and not a bacterium. Names you can safely quote as examples are Melissa, ILOVEYOU and WannaCry. If you remember an older textbook name such as Chernobyl, that is also acceptable. Give the definition first, then two examples.

Malware	How it differs
Virus	Needs a host file. Spreads when the host runs or is shared.
Worm	A standalone program that spreads across a network by itself, without a host file.
Trojan horse	Disguised as useful software. It does not have to copy itself. The user is tricked into running it.

An **antivirus** is a utility that detects, blocks and removes known malware, usually by comparing files with a list of signatures and by watching suspicious behaviour. Five names are enough for the short question: Windows Defender, Avast, AVG, Kaspersky and Bitdefender. Norton and McAfee are equally acceptable. The product must be kept updated, because a list from last year does not know this month's virus. Antivirus does not replace careful habits: do not open an unexpected attachment, and do not install cracked software.

Full forms for the “any three” question

Section B often says “write the full form of any three”. Write the full words, not a second abbreviation. These are the ones that actually appear.

Short form	Full form
OCR	Optical Character Recognition
OMR	Optical Mark Recognition (or Optical Mark Reader)
GUI	Graphical User Interface
SVGA	Super Video Graphics Array
CRT	Cathode Ray Tube
BIOS	Basic Input Output System
HTML	Hypertext Markup Language
SMTP	Simple Mail Transfer Protocol
HTTP	Hypertext Transfer Protocol
FTP	File Transfer Protocol
URL	Uniform Resource Locator
USB	Universal Serial Bus
LAN / MAN / WAN	Local / Metropolitan / Wide Area Network
OSI	Open Systems Interconnection
ISO	International Organization for Standardization
CPU / ALU / CU	Central Processing Unit / Arithmetic and Logic Unit / Control Unit
RAM / ROM	Random Access Memory / Read Only Memory
NIC	Network Interface Card
MODEM	Modulator–Demodulator
IT	Information Technology

Copyright, in one paragraph if it is asked: software is owned by its author. A licence is the permission to use it. Copying a program and giving it out without that permission is piracy, and it is illegal. Data that identifies a person should not be collected or published without a proper reason. That is the privacy side of the same short question.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The markup language used to produce web pages is:

- A) SMTP
- B) HTML

- C) FTP
- D) TCP

2. The most common protocol for sending email is:

- A) FTP
- B) HTML
- C) SMTP
- D) OCR

3. A set of rules for communication is a:

- A) Topology
- B) Protocol
- C) Plotter
- D) Register

Short answers

S1. Define a computer virus and name any five antivirus programs.

S2. Write the full form of any three: OCR, GUI, SVGA, CRT, BIOS.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. SMTP carries email. HTML describes the page.

MCQ 2. C. Simple Mail Transfer Protocol.

MCQ 3. B. Both sides must follow the same rules.

S1. A computer virus is a program that attaches to a host file, spreads when that host is shared or run, and may damage data or waste resources. Five antivirus programs are Windows Defender, Avast, AVG, Kaspersky and Bitdefender. An antivirus must be updated so that it recognises new viruses.

S2. OCR is Optical Character Recognition. GUI is Graphical User Interface. SVGA is Super Video Graphics Array. CRT is Cathode Ray Tube. BIOS is Basic Input Output System. Any three of these full forms satisfy the question.